Exercise 0.1. (2 points) Let $\phi \equiv \Big[\forall x(P(x) \vee Q(x))\Big] \rightarrow \Big[\neg(P(f(x,y)) \vee Q(y))\Big]$ and $\psi \equiv \forall x \Big[(P(x) \wedge Q(x)) \rightarrow \neg(P(f(x,y) \vee Q(y))\Big]$.

(i) Draw parse trees and identify the free and bound variables for ϕ and ψ.

(ii) Determine $\phi[(g(y)/x)]$ and $\psi[z/y]$.

Exercise 0.2. (2 points) Verify the following sequents using natural deduction rules.

(i) $\exists x P(x), \forall x \forall y(P(x) \rightarrow Q(y)) \vdash \forall y Q(y)$;

(ii) $\forall x(P(x) \wedge Q(x)) \vdash \forall x P(x) \wedge \forall x Q(x)$.

Exercise 0.3. (2 points) For each of the formula/sequent below, find a model that satisfies the formula/sequent and one that does not.

(i) $\exists x(P(x) \rightarrow S) \vdash (\exists x P(x)) \rightarrow S$;

(ii) $\forall x \Big[P(f(x)) \rightarrow R(x,a)\Big]$

Exercise 0.4. (2 points) Which of the following semantic entailments hold?

(i) $(\forall x P(x)) \rightarrow (\forall x Q(x)) \vDash \forall x(P(x) \rightarrow Q(x))$;

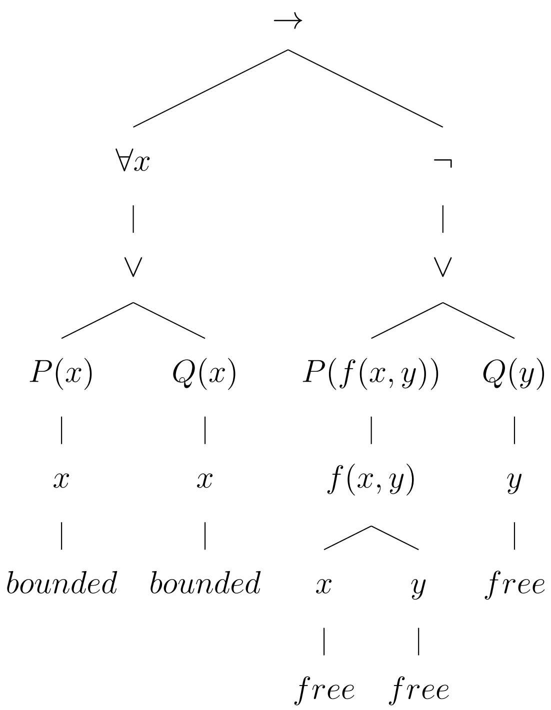
(ii) $\forall x \exists y(P(x) \vee Q(y)) \vDash \exists y \forall x(P(x) \vee Q(y))$

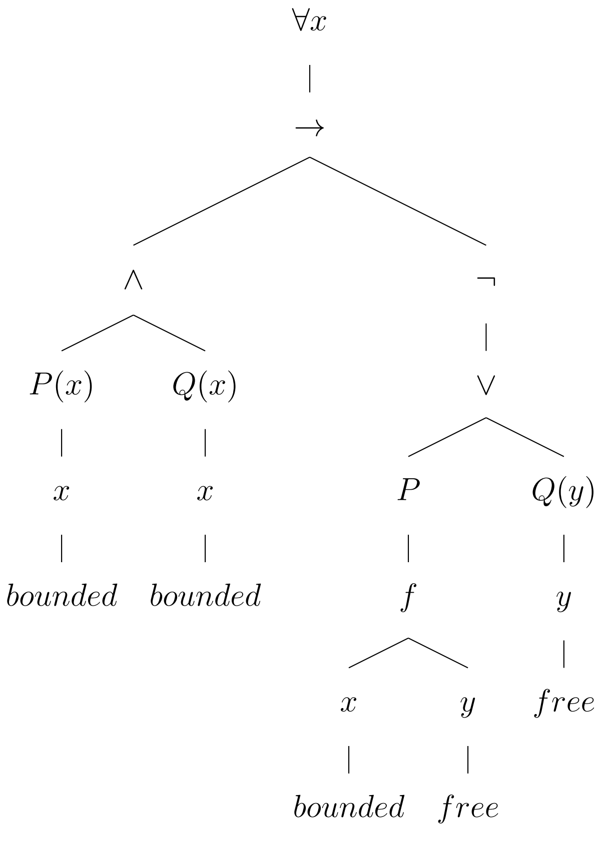
Exercise 0.5. (2 points) Use the tableau algorithm to determine whether the following formulas is valid or not. If it's not valid, is it satisfiable?

(i) $\phi \equiv \Big[(p \rightarrow r) \vee (q \rightarrow r)\Big] \rightarrow \Big[(p \wedge q) \rightarrow r\Big]$

(ii) $\psi \equiv \forall x(P(x) \vee Q(x)) \rightarrow (\forall x P(x) \vee \forall x Q(x))$

Exercise 0.6 (Exercise 0.1)**.** (a)





(b) Replacing the free variable x with $g(y)$ in ϕ, we have

$$\phi \equiv \Big[\forall x(P(x) \vee Q(x))\Big] \to \Big[\neg(P(f(g(y),y)) \vee Q(y))\Big] .$$

Replacing the free variable y with z in ψ, we have

$$\psi \equiv \forall x \Big[(P(x) \wedge Q(x)) \to \neg(P(f(x,z) \vee Q(z))\Big] .$$

Exercise 0.7 (Exercise 0.2)**.** (a) Verify $\exists x P(x), \forall x \forall y (P(x) \to Q(y)) \vdash \forall y Q(y)$

1.	$\exists x P(x)$	premise	
2.	$\forall x \forall y (P(x) \to Q(y))$	premise	
3.	x_0	assume from $\exists$E	(1)
4.	y_0	arbitrary	
5.	$\forall y (P(x_0) \to Q(y))$	$\forall E$	(2)
6.	$P(x_0) \to Q(y_0)$	$\forall E$	(4)
7.	$P(x_0)$	assumption (from (3))	
8.	$Q(y_0)$	$\to$E	(6,7)
9.	$\forall y Q(y)$	$\forall I$	(4–8)
10.	$\forall y Q(y)$	$\exists E$	(1,3–9)

(b) Verify $\forall x (P(x) \wedge Q(x)) \vdash \forall x P(x) \wedge \forall x Q(x)$

1.	$\forall x(P(x) \wedge Q(x))$	premise	
2.	x_0	arbitrary	
3.	$P(x_0) \;\wedge\; Q(x_0)$	$\forall E$	(1)
4.	$P(x_0)$	$\wedge_1$E	(3)
5.	$\forall x P(x)$	$\forall I$	(2–4)
6.	x_0	arbitrary	
7.	$P(x_0) \;\wedge\; Q(x_0)$	$\forall E$	(1)
8.	$Q(x_0)$	$\wedge_2$E	(3)
9.	$\forall x Q(x)$	$\forall I$	(6–8)
10.	$\forall x P(x) \wedge \forall x Q(x)$	$\wedge$I	(5,9)

Exercise 0.8 (Exercise 0.3). (a) $\exists x(P(x) \to S) \vdash (\exists x P(x)) \to S$

Model that satisfies:

- Domain $D = \{1, 2\}$
- $P(1) =$ false, $P(2) =$ false
- $S =$ true

  Explanation: $\exists x(P(x) \to S)$ is true because $P(1) \to S =$ true (false implies true), and $(\exists x P(x)) \to S$ is also true since $\exists x P(x) =$ false.

Model that does not satisfy:

- Domain $D = \{1, 2\}$
- $P(1) =$ true, $P(2) =$ false
- $S =$ false

  Explanation: $\exists x(P(x) \to S)$ is true because $x = 2$ satisfies $P(2) \to S =$ true, but $(\exists x P(x)) \to S =$ false $\to$ false $=$ false, so the sequent fails.

(b) $\forall x \left[P(f(x)) \to R(x, a)\right]$

Model that satisfies:

- Domain $D = \{1, 2\}$
- $f(x) = x$, $P(x) =$ false for all x
- $R(x, a)$ arbitrary

  Explanation: For all x, $P(f(x)) =$ false, so the implication is vacuously true.

Model that does not satisfy:

- Domain $D = \{1, 2\}$
- $f(1) = 2$, $P(2) =$ true, but $R(1, a) =$ false

  Explanation: Then $P(f(1)) = P(2) =$ true and $R(1, a) =$ false, so the implication fails at $x = 1$.

Exercise 0.9 (Exercise 0.4)**.** (a) $(\forall x\, P(x)) \to (\forall x\, Q(x)) \vDash \forall x\, (P(x) \to Q(x))$

Counterexample: Conside M with look up table l

- Domain $D = \{1, 2\}$
- $P(1) =$ true, $P(2) =$ false
- $Q(1) =$ false, $Q(2) =$ true

  Explanation:

  Since $P(2)$ =false, so $(\forall x\, P(x))$ is false. Then, $M_l \vDash (\forall x P(x)) \to (\forall x Q(x))$ does hold. However, when $x = 1$, we conclude $P(1)$ =true and $Q(1)$ =false. This implies $M_{l[x \to 1]} \vDash (P(x) \to Q(x))$ does not hold. Therefore $M_l \vDash \forall x (P(x) \to Q(x))$ does not hold, this implies that model M does not satisfy the formula.

(b) $\forall x\, \exists y\, (P(x) \vee Q(y)) \vDash \exists y\, \forall x\, (P(x) \vee Q(y))$.

Let M be any model with look up table l such that

$$M \vDash \forall x \exists y (P(x) \vee Q(y)) \tag{1}$$

holds. We will consider two following cases:

- There is element y_0 in D (domain) such that $M \vDash_{l[y\to y_0]} Q(y)$ holds. Hence, for any $x_0 \in D$, then $M \vDash_{l[x\to x_0, y\to y_0]} (P(x) \vee Q(y))$ holds. Therefore $\exists y \forall x (P(x) \vee Q(y))$ holds.
- For any $y_0 \in D$ such that $M \vDash_{l[y\to y_0]} Q(y)$ does not holds. Then, from (1), for an arbitrary $x_0 \in D$, we have $M \vDash_{l[x\to x_0]} \exists y (P(x) \vee Q(y))$ holds. Then, there is a $y_0 \in D$ such that

$$M \vDash_{l[x\to x_0],[y\to y_0]} (P(x) \vee Q(y)).$$

  However, $M \vDash_{l[y\to y_0]} Q(y)$ does not holds for any $y_0 \in D$. Hence, $M \vDash_{l[x\to x_0]} (P(x)$ holds for any $x \in D$. Therefore $M \vDash_l \forall x P(x)$ holds. Thus, $M \vDash_l \exists y \, \forall x \, (P(x) \vee Q(y))$ holds.

In both cases, we can conclude that this semantic entailment holds.

Exercise 0.10 (Exercise 0.5)**.** (a) $\phi \equiv \big[(p \to r) \vee (q \to r)\big] \to \big[(p \wedge q) \to r\big]$

To check validity, we negate the formula and construct the tableau for:

$$\neg \Big(\big[(p \to r) \vee (q \to r)\big] \to \big[(p \wedge q) \to r\big] \Big)$$

which is equivalent to:

$$\big[(p \to r) \vee (q \to r)\big] \wedge \neg \big[(p \wedge q) \to r\big]$$

The negation becomes:

$$\phi_1 = \big[(\neg p \vee r) \vee (\neg q \vee r)\big] \wedge (p \wedge q \wedge \neg r)$$

$$
\begin{tikzcd}[column sep=small]
 & \phi_1 = \Big[(\neg p \vee r) \vee (\neg q \vee r)\Big] \wedge (p \wedge q \wedge \neg r) \arrow[d, "\wedge - rule"] & \\
 & \phi_2 = \phi_1 \cup \{(\neg p \vee r) \vee (\neg q \vee r), p \wedge q \wedge \neg r\} \arrow[d, "\wedge - rule (2\ times)"] & \\
 & \phi_3 = \phi_2 \cup \{p, q, \neg r\} \arrow[d, "\vee - rule \text{ for } (\neg p \vee \neg q \vee r)\text{, we branch here}"] & \\
 & {} \arrow[dl] \arrow[dr] & \\
\phi_4 = \phi_3 \cup \{\neg p \vee \neg q\} & & \phi_5 = \phi_3 \cup \{r\}
\end{tikzcd}
$$

Continuing by applying the $\vee$-rule, we see that each branch leads to a contradictory set: $\{\neg p, p\}; \{q, \neg q\}$ and $\{\neg r, r\}$. Therefore, all branches close. This implies $\neg\phi$ is not satisfiable. Hence, the formula ϕ is valid.

(b) $\psi \equiv \forall x(P(x) \vee Q(x)) \to (\forall x P(x) \vee \forall x Q(x))$

To check validity, we negate the formula:

$$\neg \Big[\forall x(P(x) \vee Q(x)) \to (\forall x P(x) \vee \forall x Q(x))\Big]$$

which becomes:

$$\forall x(P(x) \vee Q(x)) \wedge \neg(\forall x P(x) \vee \forall x Q(x))$$

Negating the disjunction gives:

$$\forall x(P(x) \vee Q(x)) \wedge (\neg\forall x P(x) \wedge \neg\forall x Q(x))$$

Equivalent to:

$$\psi_1 = \forall x(P(x) \vee Q(x)) \wedge (\exists x \neg P(x) \wedge \exists x \neg Q(x))$$

$$
\begin{array}{c}
\psi_1 = \forall x(P(x) \vee Q(x)) \wedge (\exists x \neg P(x) \wedge \exists x \neg Q(x)) \\
\Big\downarrow {\scriptstyle \wedge-rule} \\
\psi_2 = \psi_1 \cup \{\forall x(P(x) \vee Q(x)), \exists x \neg P(x) \wedge \exists x \neg Q(x)\} \\
\Big\downarrow {\scriptstyle \wedge-rule} \\
\psi_3 = \psi_2 \cup \{\exists x \neg P(x), \exists x \neg Q(x)\} \\
\Big\downarrow {\scriptstyle \exists-rule} \\
\psi_4 = \psi_3 \cup \{\neg P(a), \neg Q(b)\} \\
\Big\downarrow {\scriptstyle \forall-rule \text{ for } (\forall(P(x) \vee Q(x)} \\
\psi_5 = \psi_4 \cup \{P(t) \vee Q(t)\} \\
\Big\downarrow {\scriptstyle \vee-rule} \\
\psi_6 = \psi_5 \cup \{P(t), Q(t)\}.
\end{array}
$$

Then, the sets $\{P(t), \neg P(a), \neg Q(b)\}$ and $\{Q(t), \neg P(a), \neg Q(b)\}$ are not closed (since $a \neq t$). Hence, ψ is not valid, but it is satisfiable when $t = b$.